RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra)
Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda
Designation / Role: Software & Data Engineer
Task Order Reference: 8 October 2026 – Pandas Telemetry Processing

2. DAILY LOG ENTRY FORM

- **Date:** 08 October 2026

- **1. Assigned Task:**

  Pandas-based telemetry ingestion, time-indexing, filtering, validation and processing

- **2. What Was Studied:**

  Studied Pandas as a Python library for structured telemetry and offline data processing. Learned how to create and work with Pandas DataFrames, organize telemetry records into rows and columns, and handle timestamp information. Studied how to convert timestamp values into a proper datetime format and use the timestamp as a DatetimeIndex for time-based data selection. . Finally, studied how processed telemetry data can be exported to CSV for further offline analysis and reporting

- **3. What Was Implemented:**

  Created a structured telemetry dataset and loaded it into a Pandas DataFrame. Converted the timestamp column into datetime format. Performed time-range selection, sorting, statistical analysis, missing-value checking, duplicate checking, and range validation for the numerical telemetry fields. Verified the final DataFrame shape with zero missing values and zero duplicate records. Exported the processed telemetry data to processed_telemetry.csv for offline use.

- **4. Tools / Libraries Used:**

  VS Code, PowerShell, Python 3.10 , Jupyter Notebook Python .venv , Pandas

- **5. Output File Name-** suraj_rudra_task/Day4/codes/pandas_telemetry_processing

  Path-https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**

  1. While installing Pandas, the command was run but the Python interactive shell opened instead of completing the intended installation workflow
  2. The date and time values in the CSV telemetry data were initially entered/represented in an incorrect format, which caused problems during timestamp-based processing.

- **7. How Problem Was Resolved:**

  1. I exited the Python interactive shell and ran the required command again from the terminal, after which Pandas was installed and imported successfully
  2. The date and time values in the CSV telemetry data were initially entered/represented in an incorrect format, which caused problems during timestamp-based processing.

- **8. Next Task Scheduled:**

  (09 October 2026)- Data validation ,boundary limits, and corrupted packet detection

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  Date: 08/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

[ ] Approved / Completed

[ ] Needs Revision / Delayed

[ ] Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026